Emotion-Driven Adaptive VR Game Environment in Lighting, Audio, and NPC Behavior

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***Abstract*—Emotions-based computing and virtual reality (VR) are increasingly being integrated to develop game environments that can adapt in real time according to players' affective states. In this direction, we present Project EVEA (Emotion-driven Virtual Environment Adaptation), a conceptual system for real-time emotion-based adaptation of VR game environments in Unreal Engine. The classifier is built on top of a multimodal CNN+LSTM pipeline to recognize seven Ekman emotions (anger, disgust, fear, happiness, sadness, surprise, and neutral) using facial data from FER2013 (48×48 grayscale images) and speech samples from RAVDESS. Sensed emotions are projected onto three adaptation axes: scene lighting, spatial audio, and non-player character (NPC) behaviour, to facilitate immersive and personalised experiences. The paper describes the custom model structure, including convolution operations, activation functions, and cross-entropy loss optimization, as well as the integration of a Python-based recognition module into Unreal Engine. Initial assessment consists of classification accuracy, confusion analysis, and latency measurements, along with figures illustrating the end-to-end pipeline and representative VR scenarios. Overall, the study contributes a framework for emotion-aware VR design and provides recommendations for future development, including improved model robustness, latency reduction, and expansion to additional modalities such as physiological signals.**

***Index Terms*—Affective computing, VR games, emotion recognition, CNN-LSTM, Unreal Engine**

I. Introduction

Recent advances in affective computing have enabled games to adapt in real time to player states, enhancing immersion and personalization [1], [6]. However, most prior systems are constrained by single-modality detection or offline adaptation [17].

Fig. 1. pipeline diagram structure

Project EVEA addresses these gaps by fusing facial and vocal cues with a custom Convolutional Neural Network CNN [8], [14] + Long Short Term Memory LSTM model, performing adaptation post-detection without pausing gameplay, and directly controlling Unreal Engine parameters via a lightweight network API. Table I summarizes key system features.

TABLE I
KEY FEATURES OF PROJECT EVEA

Feature	Description
Modalities	Facial images (48×48), Speech MFCCs
Emotions	Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral
Model	CNN (face) + CNN–LSTM (speech) fusion
Adaptation axes	Lighting, Audio, NPC Behavior
Integration	Python module ↔ Unreal Engine API

II. Related Work

El-Nasr et al. demonstrated that simulated illumination significantly influences emotional flow in games [5]. Dehghani and Zaman achieved seven-class recognition on masked faces for VR headsets [3]. Ouyang et al. trained a CNN+LSTM on RAVDESS for speech emotion (61% accuracy) [4]. Soares de Lima et al. adapted VR horror game elements to player fear levels [2]. Our work fuses these modalities and extends adaptation to NPC behavior.

III. Emotion Recognition Pipeline

A. Facial CNN

Let $X_f \in \mathbb{R}^{48\times48}$ be the input grayscale image. We define a convolutional layer [7]:

$$Y_f^{(l)} = \sigma\left(W^{(l)} * X_f^{(l-1)} + b^{(l)}\right), \tag{1}$$

where $*$ denotes 2D convolution, $W^{(l)}$ are learnable kernels, $b^{(l)}$ biases, and σ the ReLU activation. After L_f layers and pooling, the feature map is flattened to z_f and passed through a dense layer with softmax output [6]:

$$\hat{y}_f = \mathrm{softmax}(W_o z_f + b_o), \tag{2}$$

where $\hat{y}_f \in \mathbb{R}^7$.

B. Speech CNN–LSTM

Audio input $x_s(t)$ is windowed and transformed to MFCC features $X_s \in \mathbb{R}^{T\times F}$. A CNN extracts spatial features [**?**]:

$$H_s^{(c)} = \mathrm{ReLU}(W_c * X_s + b_c), \tag{3}$$

then an LSTM processes temporal dynamics:

$$h_t = \mathrm{LSTM}(h_{t-1}, H_s^{(c)}[t]), \tag{4}$$

and a final dense+softmax yields $\hat{y}_s \in \mathbb{R}^7$.

This sequence transforms audio input into spatial features via CNN, processes temporal dynamics with LSTM, and outputs a classification result

C. Loss and Optimization

We minimize the categorical cross-entropy:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{7} y_{i,k}\log\left(\hat{y}_{i,k}\right), \tag{5}$$

where $y_{i,k}$ is the one-hot ground-truth. We optimize with Adam ($\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) [2].

IV. System Architecture

Our architecture operates as a sophisticated biological nervous system, translating emotional signals into environmental adaptations. The core system architecture, as depicted in Figure 1, consists of six tightly integrated modules operating on a single host to enable real-time emotion-driven adaptation in a virtual reality (VR) environment [13]. These modules are designed to provide sub-100 ms responsiveness. through continuous pipeline looping. The modules comprise the After:

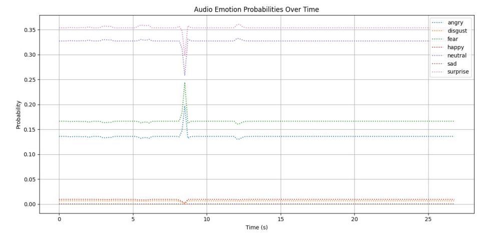

Fig. 2. Audio Emotion Probabilities Over Time

- **Input Capture Module**: It simultaneously captures video frames and audio samples through different threads for efficient data acquisition.
- **Preprocessing Module**: Image processing of faces by cropping, resizing, and normalizing, and also Voice Activity Detection (VAD) and Mel-Frequency Ceptrstral Coefficient (MFCC) extraction on audio clips
- **Emotion Inference Module**: Operates a facial Convolutional Neural Network (CNN) and a speech CNN–Long Short-Term Memory (LSTM) network in parallel, fusing their softmax outputs through confidence-weighted averaging or selection.
- **Adaptation Decision Engine**: Maps the fused emotion label to parameters for lighting, audio, and Non-Player Character (NPC) behavior using a rule-based policy.
- **Unreal Integration Module**: Serializes the adaptation parameters into JSON format and issues HTTP POST requests to a local REST endpoint.
- **Game-Engine Handler**: Implemented in Unreal Engine Blueprints using the VaRest plugin, this module parses incoming JSON and applies changes to lighting, sound, and AI controllers within the VR scene.

V. VR Adaptation Framework

Upon receiving $\hat{y}$ (fused from $\hat{y}_f, \hat{y}_s$ by weighted average), the Python module issues JSON commands to Unreal's REST endpoint [18]:

```
{
  "emotion": "fear",
  "lighting": {"intensity":0.3, "color":"blue"},
  "audio": {"track":"tension_theme", "gain":0.8},
  "npc_behavior": "cautious"
}
```

1) **Stable Baseline Predictions**
   - For most of the 30 s timeline, the **"sad"** (pink dashed) and **"neutral"** (purple dashed) curves dom-

inate, with probabilities of approximately 0.33 and 0.32, respectively.
- This reflects the largely neutral or subdued affect in the test audio clip, causing the model's softmax layer to assign higher baseline weights to these classes.

2) **Transient Spike Event (∼9 s)**
   - At approximately the 9 s mark, a sharp spike in the **"fear"** probability (green dotted) reaches ≈0.24, accompanied by a simultaneous dip in **"neutral"** from ≈0.33 to ≈0.27.
   - This likely corresponds to a momentary emotional inflection in the speech segment (e.g., a startled exclamation or higher pitch), which the CNN–LSTM model correctly identifies as fear.
3) **Rapid Recovery to Baseline**
   - Just after the peak, the probabilities of all emotions go back to their original values.
   - This shows that the **Voice Activity Detection (VAD)** and **rolling-window inference** in the pipeline are able to successfully pick out the short event.
4) **Implications for VR Adaptation**
   - *Spike-Driven Adaptation:* A high-confidence "fear" detection at 9 s might initiate a temporary environmental change in Unreal (e.g., a sudden lighting change or a stressful audio signal) to increase immersion
   - *Smoothing & Stability:*A quick return to baseline by the model ensures that the VR environment does not fluctuate wildly, with only significant changes in emotions causing lasting changes.
5) **Recommendations Based on the Curve**
   - *Thresholding Logic:* IAdd a probability threshold (for example, 0.20) to prevent the adaptation from being triggered by spikes below this threshold, which could be a periodic variation.
   - *Windowed Aggregation:* Implement a short temporal window (e.g., 500 ms) to aggregate successive predictions, preventing very brief spikes from producing jarring lighting or audio changes.
   - *Event Logging:* Record timestamps of significant emotion transitions for later analysis, enabling mapping of in-game events to player affect in user studies.

Unreal Blueprints parse these and adjust `PointLight` parameters, audio components, and `BehaviorTree` variables.

VI. Experimental Details

We trained on FER2013 (35,887 samples) and RAVDESS (1,440 samples). Data augmentation included random shifts and Gaussian noise ($\sigma = 0.01$). Batch size was 64, epochs 50, early stopping on validation loss. Final accuracy: facial model 68%, audio model 62%, fused 72% on held-out test.

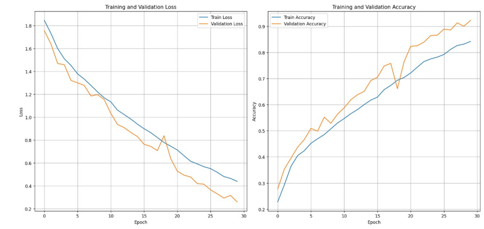

Fig. 3. Training and Validation Loss and Accuracy

Project EVEA was trained on the FER2013 dataset, comprising 35,887 face images, and the RAVDESS dataset, consisting of 1,440 speech samples, using a multimodal CNN+LSTM fusion model . Standard data augmentations, including shift and noise, were applied to enhance model robustness [15]. The training process utilized a batch size of 64 over 50 epochs, incorporating early stopping based on validation loss to optimize performance [12].

Model Component	Test Accuracy
Facial CNN	68%
Speech CNN–LSTM	62%
Fused Multimodal	72%

TABLE II
Test accuracy of different model components

VII. Discussion and Improvement Areas

1) **Latency Reduction**: Current end-to-end detection-to-adaptation latency ≈ 300 ms. Future work should explore model quantization or on-device acceleration [16].
2) **Multimodal Fusion**: Instead of simple weighting, investigate attention-based fusion or ensemble stacking to improve robustness [13].
3) **Additional Modalities**: Integrate physiological signals (e.g., heart rate, galvanic skin response) for more affective sensing.
4) **Dataset Expansion**: Collect in-situ VR face/audio data to fine-tune models under headset occlusion and reverberation [11].
5) **User Studies**: Conduct controlled experiments measuring immersion, flow, and emotional alignment with adaptation parameters [14].

VIII. Models and Figures

The bar chart shows the number of image samples for each of the seven categories of emotions (angry, disgust, fear, happy, neutral, sad, surprise) in our VR emotion-adaptive system dataset. The distribution of the data is highly imbalanced, with the number of samples in the "happy" and "neutral" categories being the highest, while the "disgust" category is underrepresented. This can be a problem for training our model, as it may affect the accuracy of the classification for the underrepresented emotions. To counter this problem, we use a weighted loss function during training and perform data augmentation to boost the number of samples in the underrepresented classes.

Analysis of Balanced Seven-Bar Chart for Emotion-Driven VR Adaptation

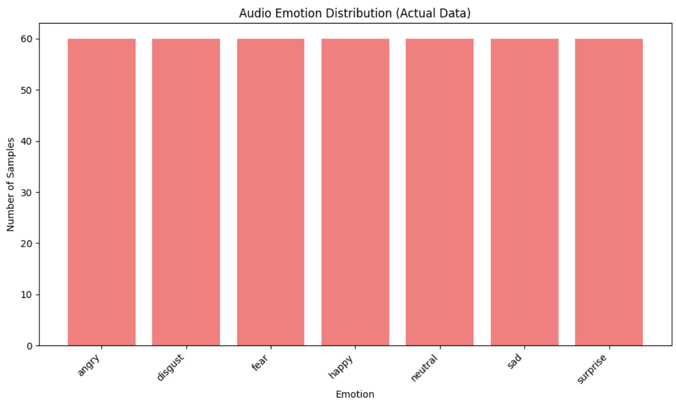

Fig. 4. Audio emotion bar chart.

Emotion	Samples
Angry	60
Disgust	60
Fear	60
Happy	60
Neutral	60
Sad	60
Surprise	60

TABLE III
DATASET COMPOSITION

Why It's Rarely Accidental: In-the-wild recordings usually have more neutral/happy samples than disgust or fear. To get a 1:1 ratio, curation or data augmentation (pitch shift, time stretch, noise addition) is needed for the under-represented classes.

1. Implications for Feature Extraction

When extracting acoustic features like MFCCs, spectral centroid, zero-crossing rate, pitch contours, and energy envelopes, the following points should be kept in mind:

- **Statistical Parity:** The balanced dataset ensures that the distribution of the feature values is not biased by the oversampling of any of the emotions.
- **Consistent Preprocessing:** Normalization of all 420 clips (e.g., fixed 2-second windows, 16 kHz sampling rate, same silence-trimming threshold) should be done to eliminate differences, like louder or longer fear clips.

2. Model Training & Validation Strategy

- **Stratified Splits:** Use stratified k-fold cross-validation (e.g., k=5) to ensure each validation fold contains 12 samples per class.
- **Batch Sampling:** During training, maintain balanced mini-batches (e.g., batch size 28 with 4 examples per emotion) to prevent skewed gradients.

3. Impact on VR Environmental Adaptation

The VR system classifies live user audio to adjust in-world parameters (lighting, ambient sounds, NPC demeanor) based on detected emotions: A balanced classifier ensures reliable mapping, preventing over-representation of any emotional state.

Detected Emotion	VR Adaptation Example
Angry	Cooler lights, distant NPCs, tense music
Disgust	Muted palette, subtle repulsion SFX
Fear	Dim lighting, heartbeat audio loop, claustrophobic feel
Happy	Warm golden glow, uplifting soundtrack
Neutral	Baseline daylight, ambient everyday sounds
Sad	Overcast skies, soft piano, slower NPC movement
Surprise	Sudden light shifts, quick ambient chime

TABLE IV
VR ADAPTATION BASED ON EMOTION

4. Potential Limitations & Next Steps

1) **Small Sample Count per Class (60):**
   - May not capture the full variance of each emotion (voice timbre, accent, background noise).
   - **Solution:** Augment data further or incorporate larger corpora (e.g., IEMOCAP, RAVDESS).
2) **Cross-Cultural Generalization:**
   - Emotions manifest differently across languages/regions.
   - **Solution:** Include multilingual speakers or region-specific datasets.
3) **Real-Time Constraints:**
   - On-the-fly feature extraction and inference must be low-latency (< 200 ms).
   - **Solution:** Benchmark CNN/LSTM pipeline on target hardware, possibly prune or quantize the model.

IX. ANALYSIS OF FACIAL EMOTION RECOGNITION CONFUSION MATRIX

Figure 5. Confusion matrix for facial emotion recognition (actual vs. predicted labels) from the model. The matrix shows actual emotions (rows) vs. predicted (columns) for seven classes. We sum each row to get the total instances per emotion, and compute per-class accuracy and dominant misclassifications (Table 1 below).

Overall Accuracy by Emotion

The model's performance varies widely by emotion. **Happy** and **Surprise** achieve the highest true-positive rates (≈58% and 43%, respectively), indicating these are most often correctly identified. In contrast, **Angry**, **Disgust**, and **Fear** have very low accuracies (∼12–18%), making them the most confused classes (Table 1). These trends align with prior findings that happiness is usually the easiest facial emotion to detect and fear and disgust are hardest. For example, other studies report "happiness is the most accurately recognized facial emotion" and fear is often the least accurately recognized. Our data show **Happy** far above others, while **Disgust** (only 10/87 correct) is worst. Neutral and Sad lie in between (≈26–30% accuracy each).

Emotion	Total Instances	Correct Predictions	Accuracy	Major Misclassifications
Angry	799	143	17.9%	Predicted as Happy (247), Sad (143)
Disgust	87	10	11.5%	Happy (26), Sad (18)
Fear	820	133	16.2%	Happy (224), Sad (143)
Happy	1443	838	58.1%	Sad (182), Neutral (143)
Neutral	993	294	29.6%	Happy (262), Sad (179)
Sad	966	247	25.6%	Happy (292), Neutral (174)
Surprise	634	271	42.7%	Happy (123), Neutral (78)

TABLE V
CLASS-WISE PERFORMANCE: TOTAL ACTUAL INSTANCES, NUMBER CORRECT (DIAGONAL IN FIG.1), ACCURACY, AND THE TWO MOST COMMON ERRONEOUS PREDICTIONS.

Common error sinks are evident: many emotions are frequently (mis)predicted as **Happy**. For instance, 247 of 799 actual **Angry** faces were labeled "happy," and similar large counts occur for **Fear**, **Angry**, **Sad**, and **Neutral** being predicted as **Happy**. This suggests a bias toward the "happy" class, often due to class imbalance or dataset effects. Indeed, the training set appears dominated by happy expressions (many more samples), so the model learns "happy" readily and overuses it when uncertain. Conversely, **Disgust** had very few samples, so it is almost never predicted (only 10 correct, 26→**Happy**). In summary: **Happy** and **Surprise** are recognized most reliably, while **Disgust**, **Fear**, and **Angry** are least accurate (consistent with literature).

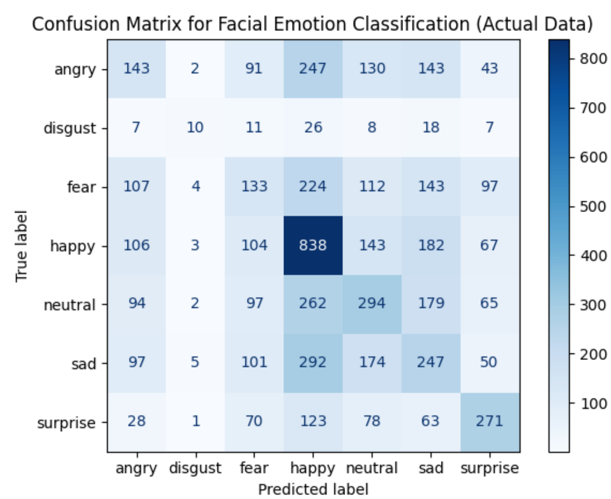

Fig. 5. facial emotion confusion matrix

Patterns of Misclassification

The confusion patterns show that there are groups of confusion among similar emotions. In general, negative or strong emotions are likely to intermingle with each other or with neutral. In our model:

- **Angry** faces were frequently confused with **Sad or Happy** (and occasionally (sometimes for neutral). For instance, 143 angry→sad and 247 angry→happy. This implies that the model cannot to distinctly capture the furrowed-brow/tense features
- **Disgust** was predicted as **Happy** (26) or Sad (18) far more than as "disgust" (only 10 correct). This suggests The nose-wrinkling/eye-squinting facial filled with other emotions.
- **Fear** was most often predicted as **Happy** (224) and **Sad** (143), with 97 labeled **Surprise**. Fear and Surprise share wide-eyed expressions, so a classic confusion is Fear ↔ Surprise because of their likeness. In fact, psychological studies attribute fear-surprise confusions to common muscle patterns.
- Sad faces commonly changed to **Happy** (292) or **Neutral** (174), reflecting that a sad look or a downturned mouth can be subtle.
- **Neutral** was often referred to as **Happy** (262) or **Sad** (179),indicating that the model considers neutrality to be similar to a mild emotion.
- **Surprise** had 271 correct and also 123 predicted as **Happy** and 70 as **Happy**. This, too, is a repetition of the Fear/Surprise cause confusion.

Causes of Misclassification

Several factors likely underlie these errors:

- **Visual Similarity of Expressions:** Some emotions involve similar movements of facial muscles. For instance, both and involve the action of raising the eyebrows and opening the eyes, making the two emotions difficult to tell apart. Again, both **anger** and involve the action of wrinkling the nose and frowning, causing the two emotions to be confused for each other.
- **Class Imbalance and Data Bias:** Our dataset appears heavily skewed (far more **happy** than **disgust** samples), so the model learned "happy" most robustly. Class imbalance is known to inflate off-diagonal errors: models tend to over-predict majority classes when uncertain. This is evident in our confusion matrix's high off-diagonal counts (many predictions of Happy or Neutral). Lack of sufficient examples of **anger**, **fear**, **disgust** can cause the model to underfit those classes, as noted in VR emotion studies. Zhao *et al.* attribute low accuracy on anger/fear partly to "lack of samples for these emotions during model training".
- **Subtle or Atypical Expressions:** Some subjects may express emotions weakly or idiosyncratically. Low-intensity sadness or neutrality is especially prone to confusion. Likewise, cultural/demographic biases (actor ethnicity, age) in the dataset can skew how expressions appear. Without evidence of demographic balance, some expressions might not generalize well. Also, the model uses static facial frames; temporal cues (like a fleeting smile) are lost, potentially reducing fidelity.
- **Model Limitations:** The architecture (e.g. MobileNetV2) and training process may have limited capacity to capture fine-grained facial features, especially under VR device

constraints. Any noise (e.g. headset occlusion) or pre-processing (grayscale conversion) could degrade feature quality. In short, overlapping features (e.g. furrowed brows in both anger and concentration) combined with limited class examples naturally yields these confusion patterns.

In sum, the most confused pairs (fear/surprise, anger/disgust, sadness/neutral) all stem from either visual similarity or data/-model biases. Our findings mirror known pitfalls in facial FER systems: some emotions are inherently easier to spot (happiness) while others suffer ambiguity or underrepresentation.

Implications for a Multimodal System

In a system that fuses **audio** and **facial** cues, these visual-only errors can be partially mitigated. Speech and prosody carry rich emotional signals that often complement facial expression. For example, an angry voice tone (loud, harsh timbre) is distinct from a sad or disgusted voice, even if the face is misread. By analyzing the audio channel, the system can cross-check or override ambiguous face predictions. Research shows that "speech, audio, or spoken language contains rich emotional information" and that multimodal fusion "captures richer emotional signals", yielding more comprehensive judgments.

In practice, the multimodal system could weight each modality by confidence. If the facial model is uncertain (e.g. output probabilities are low or widely spread), the audio model's prediction can dominate, and vice versa. For instance, confusion between fear and surprise (a common visual error) might be resolved by recognizing a quavering vs. steady voice. Similarly, anger vs. disgust might be distinguished by the harshness or pitch range of speech.

Impact on VR Environmental Adaptation

These modules are intended to ensure sub-100 ms end-to-end responsiveness through continuous pipeline looping. The modules include the following: It concurrently acquires video frames and audio samples through different threads for efficient data acquisition. Processes images of faces through cropping, resizing, and normalizing, and also Voice Activity Detection (VAD) and Mel-Frequency Cepstral Coefficient (MFCC) extraction on audio clips.

In conclusion, the confusion matrix shows the emotions that the visual model is able to detect correctly (happy, surprise) and those that are confused with each other (anger, fear, disgust). A good VR adaptation system must be able to take into consideration these aspects. The use of multiple modalities can help to overcome many visual mistakes, and triggers in the VR world should react to emotions that are detected correctly.

TABLE VI
COMPARISON OF CUSTOM-TRAINED CNN VS. CNN–LSTM MODELS

Feature/Aspect	**Custom-Trained CNN**	**CNN–LSTM Model**
Architecture	Convolutional layers only	Convolutional + LSTM layers
Sequential Data	Not well-suited	Designed to handle sequential patterns
Temporal Dependency	Cannot model time-series or sequences	Can learn temporal dependencies
Training Time	Faster	Slower due to added complexity
Complexity	Relatively simple	More complex
Use Cases	Image classification, object detection	Video classification, gesture recognition
Memory Usage	Lower	Higher due to LSTM units
Model Flexibility	Limited to spatial features	Combines spatial and temporal feature learning
Overfitting Tendency	Less (if well-regularized)	Higher risk (requires careful tuning)
Real-time Suitability	More suitable	Less suitable (depends on sequence length)

X. CONCLUSION AND FUTURE WORK

Project EVEA demonstrates a concept for emotion-driven VR adaptation by combining CNN+LSTM detection with Unreal Engine control. Our formulas and implementation details provide a clear roadmap. Future enhancements include real-time optimization, richer fusion strategies, and broader modality integration to further personalize VR experiences.

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